Question	Answer	Marks	Partial Marks
1(a)(i)	1580 or 1583 to 1584	**2**	**M1** for $\pi \times 6^2 \times 14$
1(a)(ii)	452 or 452.3 to 452.4...	**2**	**M1** for $\left[\frac{1}{2}\right] \times \frac{4}{3} \times \pi \times 6^3$
1(b)(i)	7.85 ÷ 1000 [= 0.00785]	**M1**	
1(b)(ii)	16[.0] or 15.95 to 15.99	**2**	**FT {***their* **(a)(i)** + *their* **(a)(ii)**} × 0.00785 evaluated to 3 sig fig or better **M1** for (*their* **(a)(i)** + *their* **(a)(ii)**) × 0.00785
1(c)(i)	16.2 or 16.21 to 16.23	**3**	**M2** for $\frac{2000 - 50 \times \frac{4}{3} \times \pi \times 2^3}{2000}$ [×100] or for $\frac{50 \times \frac{4}{3} \times \pi \times 2^3}{2000} \times 100$ or **M1** for $\frac{50 \times \frac{4}{3} \times \pi \times 2^3}{2000}$
1(c)(ii)	6.87 or 6.870 to 6.872	**1**	FT $\sqrt[3]{2000 - \textit{their}\left(50 \times \frac{4}{3} \times \pi \times 2^3\right)}$ evaluated to 3sf or better
1(d)	$\frac{2}{3}$ oe	**4**	**M1** for $[\pi](3R)^2 + [\pi]3R \times 9R$ oe **M1** for $2[\pi]x^2 + 2[\pi]x \times 7x$ oe **M1** for *their* area of cone = *their* area of cylinder seen
2(a)(i)	2990 cao	**1**	
2(a)(ii)	1.0 cao	**1**	
2(a)(iii)	2100 cao	**1**	
2(b)	97	**1**	
2(c)	$\frac{1}{64}$ final answer	**1**	
2(d)	$7.01[0] \times 10^{-3}$	**1**	
2(e)	1.65×10^x	**2**	**M1** for final answer figs 165 or for $15 \times 10^{x-1}$ seen or for 0.15×10^x seen

Question	Answer	Marks	Partial Marks
2(f)	37.7... – 3.7... [= 34] oe	**M1**	
	$\frac{34}{90}$ oe fraction	**B1**	
3(a)	$-2 < x \leqslant 4$ oe	**1**	
3(b)(i)	$-3 \leqslant x < 3$ final answer	**3**	**M2** for $-3 \leq x < k$ or for $k \leqslant x < 3$ or for $-6 \leqslant 2x < 6$ or for $-\frac{3}{2}-\frac{3}{2} \leqslant x < \frac{9}{2}-\frac{3}{2}$ or **M1** for $-3-3 \leqslant 2x < 9-3$ or for $-\frac{3}{2} \leqslant x + \frac{3}{2} < \frac{9}{2}$ After 0 scored SC1 for $-3 \leqslant x$ or for $x < 3$
3(b)(ii)	–3, –2, –1, 0, 1, 2 final answer	**2**	**FT** *their* **(i)** as long as negative and positive values **B1FT** for one error or omission
3(c)(i)	$\frac{36}{17}$ oe	**4**	**B3** for $-15x-2x = 5 + 4 - 45$ or better OR **B2** for $45 - 15x - 2x - 4 = 5$ oe OR **M1** for correct removal of fraction or **M1** for correct removal of brackets
3(c)(ii)	–8	**3**	**B2** for $5x - 3x = 9 - 25$ or better or **M1** for $5(x + 5) = 3(x + 3)$ oe or better
4(a)(i)	550 nfww	**3**	**M2** for $\frac{500\times2\times5}{100}+500$ oe or **M1** for $\frac{500\times2\times5}{100}$ oe
4(a)(ii)	546.65	**2**	**M1** for $500\times\left(1+\frac{1.8}{100}\right)^5$ oe
4(a)(iii)	8 nfww	**3**	**B2** for final answer 13 OR **M2** for trials correctly comparing both investments to 7 and 8 more years or **M1** for at least two trials correctly comparing both investments